

The sharp square-root bill of an exact causal source

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Abstract

Let P and Q be finite nondegenerate laws with the same Shannon entropy h and surprisal standard deviations $\sigma_P \neq \sigma_Q$. An exact horizon- n causal source is a finite seed which, under every deterministic nonanticipative policy, emits the product law of the requested actions. Its minimal seed entropy admits the second-order expansion

$$C_n^{\text{causal}}(P, Q) = nh + \sqrt{\frac{2}{\pi}} |\sigma_P - \sigma_Q| \sqrt{n} + o(\sqrt{n}).$$

The lower bound is not the two-transcript profile bound. It comes from evaluating the seed atomwise against a grid of threshold policies simultaneously: the seed mass is at most every transcript probability at once, so the seed surprisal dominates the maximum of the centered policy martingales. Those martingales converge to two-speed Brownian motions, and the lower envelope of their time-1 laws has mean exactly $\sqrt{2/\pi} |\sigma_P - \sigma_Q|$. The matching upper bound is a one-way quantile recycling of two independent streams, lifted to a history-indexed source by reading the second stream backwards. The recycled fiber length tracks the running maximum of the surprisal-difference walk, which has the same leading constant. For the crash pair $P = (1/2, 1/4, 1/4)$ and $H(Q) = 3/2$ the coefficient is $0.0347472540518177\dots$

1 Model

Let P and Q be probability laws on finite alphabets, with every atom strictly positive, and write

$$h = H(P) = H(Q), \quad \sigma_P^2 = \text{Var}(-\log_2 P(X)), \quad \sigma_Q^2 = \text{Var}(-\log_2 Q(Y)).$$

Both variances are finite. The argument is symmetric in P and Q ; from Section 3 onward we set

$$a = \sigma_P \geq b = \sigma_Q$$

and write $\Delta = a - b \geq 0$. The centered one-letter surprisals are uniformly bounded: there is a finite $L_\star$ such that

$$|-\log_2 P(x) - h| \leq L_\star, \quad |-\log_2 Q(y) - h| \leq L_\star$$

for every positive atom.

A deterministic nonanticipative policy $\pi = (\pi_t)_{t=1}^n$ chooses an action

$$A_t = \pi_t(Y_1, \dots, Y_{t-1}) \in \{P, Q\}.$$

An exact horizon- n history-indexed causal source is a finite random seed S together with deterministic output maps such that, for every such policy, the transcript Y^π has law

$$R_\pi(y^n) = \prod_{t=1}^n P_{A_t}(y_t), \tag{1}$$

where $P_P = P$ and $P_Q = Q$. Equivalently, (1) holds for every deterministic action word, and hence for every adaptive policy. The causal cost is the minimal seed entropy

$$C_n^{\text{causal}}(P, Q) = \inf \{ H(S) : S \text{ is an exact horizon-}n \text{ seed} \}.$$

For a policy π and a transcript y^n write

$$I_\pi(y^n) = -\log_2 R_\pi(y^n), \quad M_{\pi,n}(y^n) = I_\pi(y^n) - nh.$$

Thus $M_{\pi,n}(Y^\pi)$ is a centered martingale in the filtration of Y^π , with increments bounded by $L_\star$.

2 The seed sees every policy at once

The two-policy profile bound only recovers half the constant below. The sharp lower bound uses the stronger atomwise comparison.

Lemma 1 (Atomwise seed inequality). *Let S be an exact seed and let Π be any finite family of deterministic nonanticipative policies. For every atom s of mass $\gamma_s > 0$,*

$$\gamma_s \leq R_\pi(y^\pi(s)) \quad \text{for every } \pi \in \Pi,$$

where $y^\pi(s)$ is the transcript produced by π on the seed atom s . Consequently

$$H(S) \geq nh + \mathbb{E} \left[\max_{\pi \in \Pi} M_{\pi,n}(Y^\pi) \right], \quad (2)$$

the expectation being under the seed law. In particular

$$C_n^{\text{causal}}(P, Q) \geq nh + \sup_{\Pi} \mathbb{E} \left[\max_{\pi \in \Pi} M_{\pi,n} \right].$$

Proof. The output maps are deterministic, so each policy transcript is a function of S . Exactness says that the push-forward of the seed law under $s \mapsto y^\pi(s)$ equals R_π . Therefore, for each transcript y ,

$$\sum_{s: y^\pi(s)=y} \gamma_s = R_\pi(y).$$

Every summand is nonnegative, so $\gamma_s \leq R_\pi(y^\pi(s))$. The same atom s satisfies the inequality for every $\pi \in \Pi$ simultaneously, hence

$$-\log_2 \gamma_s \geq \max_{\pi \in \Pi} I_\pi(y^\pi(s)) = nh + \max_{\pi \in \Pi} M_{\pi,n}(y^\pi(s)).$$

Averaging over s gives (2). □

The inequality is strictly stronger than a bound that uses only marginal transcript laws. The profile functional of the companion note,

$$\mathcal{L}(Z_{\pi_1}, \dots, Z_{\pi_m}) = \int_0^\infty \max_i \mathbb{P}(Z_{\pi_i} > t) dt,$$

depends on the marginal tails alone. Lemma 1 charges the joint maximum realized by one seed. Azuma's inequality still shows that each $M_{\pi,n}$ is $O(\sqrt{n})$ in L^1 , uniformly in the policy, so the gain available here is a constant factor inside the square-root scale, not a change of exponent.

For two policies the profile functional collapses to the familiar Wasserstein form, recorded only for comparison.

Proposition 2 (Two-policy floor). *For the two constant policies,*

$$C_n^{\text{causal}}(P, Q) \geq nh + \frac{1}{2}W_1(S_{P,n}, S_{Q,n}),$$

where $S_{P,n}$ and $S_{Q,n}$ are the product surprisal sums. Hence

$$\liminf_{n \rightarrow \infty} \frac{C_n^{\text{causal}}(P, Q) - nh}{\sqrt{n}} \geq \frac{|\sigma_P - \sigma_Q|}{\sqrt{2\pi}}.$$

Proof. Lemma 1 with the two constant policies yields $H(S) \geq \mathbb{E} \max\{S_{P,n}, S_{Q,n}\}$. For any coupling of two nonnegative random variables, $\mathbb{E} \max(U, V) \geq (\mathbb{E}U + \mathbb{E}V + W_1(\mathcal{L}(U), \mathcal{L}(V)))/2$, and both transcript entropies equal nh . Finite support gives W_1 convergence of the normalized sums to the centered Gaussians of variances σ_P^2 and σ_Q^2 . The monotone coupling of those Gaussians has cost $|\sigma_P - \sigma_Q| \mathbb{E}|G| = \sqrt{2/\pi} |\sigma_P - \sigma_Q|$, and the factor $1/2$ produces the stated liminf. $\square$

The constant in Proposition 2 is exactly half the constant in Theorem 11. The rest of the lower bound is the recovery of that factor two.

3 Threshold policies

Fix $z \in \mathbb{R}$. The threshold policy $\pi_{z,n}$ requests the high-variance action while the centered surprisal accumulated so far lies below the barrier, and the low-variance action afterwards:

$$A_t = \begin{cases} P & \text{if } M_{t-1} < z\sqrt{n}, \\ Q & \text{if } M_{t-1} \geq z\sqrt{n}. \end{cases}$$

Write $M_n^{(z)}$ for its terminal centered surprisal and $X^{(z)}$ for the diffusion

$$dX_t = \sigma_z(X_t) dB_t, \quad X_0 = 0, \quad \sigma_z(x) = \begin{cases} a & \text{if } x < z, \\ b & \text{if } x \geq z. \end{cases} \quad (3)$$

The diffusion coefficient is Borel, bounded, and uniformly elliptic. The martingale problem for (3) is well posed, and the time-1 law is absolutely continuous [3, 1]. Call this time-1 law F_z .

Lemma 3 (Diffusion limit of one threshold). *For each fixed z ,*

$$\frac{M_n^{(z)}}{\sqrt{n}} \Rightarrow X_1^{(z)}.$$

The convergence holds jointly for any finite set of thresholds. Moreover the family $\{|M_n^{(z)}|/\sqrt{n}\}$ is uniformly integrable, uniformly in z and n .

Proof. The centered surprisal is a martingale with bounded increments. Conditional on the past, the next increment has mean zero and variance a^2 or b^2 according to the side of the barrier $z\sqrt{n}$ on which the current position lies; the conditional variance equals the exact one-letter varentropy of the requested action. Standard functional convergence for triangular martingale arrays with bounded increments and state-dependent conditional variance applies: the normalized path converges weakly in Skorokhod space to the unique solution of the martingale problem (3). See Ethier–Kurtz, Chapter 7, for the martingale-problem criterion, and Lejay for uniqueness at a single discontinuous interface. Joint convergence on a finite grid follows by applying the same criterion to the vector of

policies driven by a common seed, or, for the marginal envelope argument below, is unnecessary: only marginal convergence is required.

Uniform integrability is Azuma's inequality. The increments are bounded by L_* , so for every policy and every $s > 0$,

$$\mathbb{P}(|M_{\pi,n}| > s) \leq 2 \exp\left(-\frac{s^2}{2nL_*^2}\right).$$

Hence $\mathbb{E}[|M_{\pi,n}|/\sqrt{n}]^{1+\delta}$ is bounded for every $\delta \in (0, 1)$, uniformly in the policy. $\square$

4 The envelope

Let Φ be the standard normal distribution function. Define

$$F_*(x) = \begin{cases} \frac{2b}{a+b} \Phi(x/b) & \text{if } x \leq 0, \\ 1 - \frac{2a}{a+b} (1 - \Phi(x/a)) & \text{if } x > 0. \end{cases} \quad (4)$$

This is a distribution function. Its mean is the constant we need, and the computation is elementary once (4) is granted.

Lemma 4 (Mean of the envelope). *If R_* has law F_* , then*

$$\mathbb{E}[R_*] = \sqrt{\frac{2}{\pi}} (a - b).$$

Proof. The Gaussian tail integrals

$$\int_0^\infty (1 - \Phi(x/c)) dx = \frac{c}{\sqrt{2\pi}}, \quad c > 0,$$

give

$$\begin{aligned} \mathbb{E}[R_*] &= \int_0^\infty (1 - F_*(x)) dx - \int_{-\infty}^0 F_*(x) dx \\ &= \frac{2a}{a+b} \cdot \frac{a}{\sqrt{2\pi}} - \frac{2b}{a+b} \cdot \frac{b}{\sqrt{2\pi}} = \frac{2}{\sqrt{2\pi}} (a - b) = \sqrt{\frac{2}{\pi}} (a - b). \end{aligned}$$

The identity $a^2 - b^2 = (a - b)(a + b)$ is the only cancellation. $\square$

The distribution F_* is not the law of a single two-speed path. It is the lower envelope of the family $\{F_z\}$, attained by placing the interface at the level being tested.

Lemma 5 (Interface formula). *For every $x \in \mathbb{R}$,*

$$F_*(x) = \mathbb{P}(X_1^{(x)} \leq x) = \inf_{z \in \mathbb{R}} \mathbb{P}(X_1^{(z)} \leq x).$$

In particular $F_(x) = \inf_z F_z(x)$.*

Proof. Write $u(t, y) = \mathbb{P}(X_1 \leq x \mid X_t = y)$ for a controlled diffusion whose instantaneous volatility σ_t is progressively measurable and takes values in $\{a, b\}$. Then $u(1, y) = \mathbf{1}_{y \leq x}$, and on $[0, 1)$ the function u satisfies the HJB equation

$$\partial_t u + \frac{1}{2} \sup_{\sigma \in \{a, b\}} \sigma^2 \partial_{yy} u = 0 \quad (5)$$

in the viscosity sense, because the controller who wants to minimize $\mathbb{P}(X_1 \leq x)$ chooses the volatility that the forward equation charges with a minus sign; equivalently, the probability of the bad half-line is the expectation of the indicator under the controlled law, and the generator contribution of a pure diffusion is $\frac{1}{2} \sigma^2 \partial_{yy}$. The candidate feedback

$$\sigma^*(y) = a \cdot \mathbf{1}_{y < x} + b \cdot \mathbf{1}_{y \geq x}$$

is exactly π_x in the diffusion limit. Its occupation formula is classical for oscillating Brownian motion after a translation that moves the interface to the origin [2, 3]. Translated back, the sub-probability density of $X_1^{(x)}$ on $(-\infty, x]$ is the left-hand Gaussian density of variance b^2 , weighted by the transmission factor $2b/(a+b)$ when the starting point lies on the low-volatility side of the interface, and the right-hand tail is the Gaussian density of variance a^2 , weighted by $2a/(a+b)$. Those are the two lines of (4), so the candidate achieves F_* .

To see that no other interface does better, observe that F_* is a viscosity solution of (5) with the same terminal condition. On $\{y < x\}$ the candidate value function is a mixture of Gaussians whose second derivative has the sign that selects $\sigma = a$; on $\{y > x\}$ it selects $\sigma = b$. Uniqueness of bounded viscosity solutions of (5) gives optimality of the threshold $z = x$. Thus the pointwise infimum over deterministic thresholds, and even over all progressively measurable volatility choices, equals $F_*(x)$. $\square$

At the single point $x = 0$ the formula reduces to the transmission weight

$$\mathbb{P}(X_1^{(0)} \leq 0) = \frac{b}{a+b},$$

which is the standard left-mass of oscillating Brownian motion started on the interface. For $x \neq 0$ the starting point is not on the interface, and the tail scale is the volatility of the side that contains the origin: scale b for $x < 0$, scale a for $x > 0$.

Lemma 6 (Envelope lower bound). *Let Π_n be a finite $n^{-1/4}$ -grid of thresholds in $[-\log n, \log n]$. Then*

$$\liminf_{n \rightarrow \infty} \mathbb{E} \left[\max_{\pi \in \Pi_n} \frac{M_{\pi, n}}{\sqrt{n}} \right] \geq \sqrt{\frac{2}{\pi}} (a - b).$$

Proof. Write $M_n = \max_{\pi \in \Pi_n} M_{\pi, n}/\sqrt{n}$. For every x ,

$$\mathbb{P}(M_n \leq x) \leq \min_{\pi \in \Pi_n} \mathbb{P}(M_{\pi, n}/\sqrt{n} \leq x).$$

Pass to the limit along the grid. Lemma 3 gives marginal convergence at every fixed threshold. The map $z \mapsto F_z(x)$ is continuous in z at continuity points of F_z , because the diffusion coefficient depends continuously on the interface in the bounded-Lipschitz metric on time-1 laws: moving the interface by η changes the volatility on an interval of length η , and the resulting total-variation distance is $O(\eta^{1/2})$ by the Burkholder–Davis–Gundy inequality. A grid of spacing $n^{-1/4}$ is therefore dense enough that

$$\limsup_{n \rightarrow \infty} \mathbb{P}(M_n \leq x) \leq \inf_z F_z(x) = F_*(x)$$

at every continuity point of F_* . Layer-cake representation and Fatou's lemma on the positive part give

$$\liminf_{n \rightarrow \infty} \mathbb{E}[(M_n)_+] \geq \int_0^\infty (1 - F_*(t)) dt.$$

For the negative part the inequality runs the other way and produces a matching limsup,

$$\limsup_{n \rightarrow \infty} \mathbb{E}[(M_n)_-] \leq \int_{-\infty}^0 F_*(t) dt,$$

after truncating at $\log n$ and using the Azuma tail to show that the truncation costs $o(1)$. Thresholds outside $[-\log n, \log n]$ do not contribute to the envelope at the $\sqrt{n}$ scale: a barrier beyond $\log n$ standard deviations is not hit with probability $1 - o(n^{-K})$ for every K , again by Azuma, so the grid restriction is free. Combining the two sides and applying Lemma 4 yields the claim. Uniform integrability of the maximum over the grid follows from the same Azuma tail and a union bound: the grid has cardinality $O(n^{1/4} \log n)$, which does not damage the Gaussian tail. $\square$

5 Matching upper bound

The construction has two layers. First produce a one-way joint of two length- n strings whose conditional entropy meets the constant. Then lift that joint to an exact history-indexed seed by reading the second string backwards.

5.1 One-way recycling

Let $L = \lfloor (\log n)^2 \rfloor$ and $m = \lfloor n/L \rfloor$, and write $n = mL + r$ with $0 \leq r < L$. The remainder r contributes $O(L) = o(\sqrt{n})$ bits and is ignored after the first sentence of the entropy budget.

Draw $W^n \sim Q^n$. Generate X^n from W^n and an auxiliary uniform seed by blocks, causally. On block $j = 1, \dots, m$, having seen W on that block, split the current seed fiber in the proportions of $P^{\otimes L}$ and retain the atom consistent with a P -block that is quantile-coupled to the observed Q -block. Concretely, let $U_0 \sim \text{Unif}[0, 1]$ be independent of W^n , and maintain the dyadic fiber $(\lambda_j^-, \lambda_j^+)$ containing U_0 . The Q -block $W^{(j)}$ has surprisal $B_j = -\log_2 Q^{\otimes L}(W^{(j)})$. The comonotone (quantile) preimage of this block under the product law $P^{\otimes L}$ is an interval of $P^{\otimes L}$ -mass

$$\alpha_j = \Theta(2^{-(B_j - A_j)}),$$

where A_j is the P -surprisal of the selected atom; the implicit factor is the usual arithmetic-coding redundancy, at most 2 per block after the Kraft tightening of $O(1)$ bits per block recorded below. Intersecting the current fiber with that preimage multiplies its length by at most α_j . The resulting string X^n is exactly P^n -distributed, because each block is drawn from the $P^{\otimes L}$ splitting of a uniform fiber, and X_t is a function of U_0 and $W_1, \dots, W_t$ only.

Lemma 7 (Fiber length tracks the running maximum). *Let A_j, B_j be the block surprisals under the comonotone coupling, and set*

$$Z_j = (A_j - Lh) - (B_j - Lh), \quad S_k = \sum_{j=1}^k Z_j, \quad S_k^+ = \max_{0 \leq i \leq k} S_i$$

with $S_0 = 0$. The terminal fiber length λ satisfies

$$-\log_2 \lambda \leq Lh \cdot 0 + S_m^+ + G_m = S_m^+ + G_m, \quad (6)$$

where the overhead G_m obeys $G_m \leq C \log m$ almost surely for a constant C depending only on the alphabets. Consequently

$$H(X^n | W^n) \leq \mathbb{E}[S_m^+] + O(\log n).$$

Proof. The construction never needs more uniform bits than the heaviest fiber restriction encountered along the path. A negative increment $Z_j < 0$ shortens the P -preimage relative to the Q -surprisal already paid by W , and the unused portion of the fiber can be returned to the pool: the nesting is a restriction followed by a release, not a permanent multiplication by every α_j . The outstanding restriction at step k is therefore bounded by the largest partial sum of the increments Z_j , up to the integer-bit loss of an arithmetic coder. Each block contributes at most $O(1)$ bits of coder redundancy (a fixed Kraft gap coming from the finite alphabet), and the record times of S_k contribute an additional $O(\log m)$ bits to address the active interval endpoints. This is the inequality (6). The conditional entropy is at most the expected codelength of a prefix-free encoding of the fiber, which is $-\mathbb{E} \log_2 \lambda + O(1)$. $\square$

The overhead $O(\log n)$ is $o(\sqrt{n})$. The walk itself produces the constant.

Lemma 8 (Walk maximum). *Under the comonotone coupling of the two block-surprisal laws,*

$$\mathbb{E}[Z_1] = 0, \quad \frac{\mathbb{E}[Z_1^2]}{L} \rightarrow (a - b)^2,$$

and

$$\frac{\mathbb{E}[S_m^+]}{\sqrt{n}} \rightarrow \sqrt{\frac{2}{\pi}} (a - b).$$

Proof. Each block is an i.i.d. copy, so $(Z_j)_{j \leq m}$ is an i.i.d. centered triangular array. Finite alphabets give a fourth-moment bound $\mathbb{E}Z_1^4 = O(L^2)$. The normalized block sums obey a CLT, and the comonotone coupling is the W_2 -optimal coupling of the two block laws. Therefore

$$\frac{Z_1}{\sqrt{L}} \Rightarrow (a - b)G, \quad G \sim N(0, 1),$$

which is the variance claim; the fourth-moment bound upgrades convergence of second moments. Donsker's theorem lifts the array to a Brownian path on $[0, 1]$ after diffusive scaling by $\sqrt{mL} = \sqrt{n - r}$. The running-maximum functional is continuous in the Skorokhod topology on the set of paths whose maximum is not attained at a flat stretch, a set of full Wiener measure. Hence $S_m^+/\sqrt{n}$ converges in law to $(a - b) \max_{t \leq 1} B_t$. Uniform integrability follows from Doob's inequality and the fourth-moment bound. The reflected Brownian maximum has expectation

$$\mathbb{E} \left[\max_{t \leq 1} B_t \right] = \sqrt{\frac{2}{\pi}},$$

by the reflection identity $\mathbb{P}(\max_{t \leq 1} B_t \geq u) = 2\mathbb{P}(B_1 \geq u)$ for $u \geq 0$. Multiplying by $a - b$ gives the limit. $\square$

5.2 Anti-diagonal lift

It remains to turn the one-way joint into an exact causal seed without increasing the entropy by more than $o(\sqrt{n})$.

Lemma 9 (Anti-diagonal lift). *Suppose $X^n \sim P^n$, $W^n \sim Q^n$, and*

$$\mathcal{L}(X_{1:k} \mid W_{1:n}) = \mathcal{L}(X_{1:k} \mid W_{1:k}) \quad \text{for every } k \leq n. \quad (7)$$

Read W backwards: on the r -th request for P , emit X_r ; on the s -th request for Q , emit W_{n-s+1} . Every action word, and every adaptive policy, then exposes independent symbols with the requested product law. The pair (X^n, W^n) is an exact horizon- n seed, and

$$C_n^{\text{causal}}(P, Q) \leq H(X^n, W^n) = nh + H(X^n \mid W^n).$$

Proof. An action word with r letters P and s letters Q , $r + s = n$, exposes $X_1, \dots, X_r$ and $W_n, W_{n-1}, \dots, W_{n-s+1}$. The reversed block is exactly $W_{r+1}, \dots, W_n$, hence disjoint from $\{1, \dots, r\}$. Condition (7) says that $X_{1:r}$ is a function of the seed fiber and $W_{1:r}$ only, so it is independent of $W_{r+1:n}$. The marginals are the requested products by construction. An adaptive policy reveals only a prefix of some action word at each time, and the same index disjointness applies pathwise: after r requests for P and s requests for Q , the reversed reads still lie strictly above index r . Thus every policy transcript has law R_π . $\square$

No padding is required, and a padding segment would be illegal for the entropy budget: an unused middle copy of W adds a second nh and destroys the first-order term. The reversed block of a mixed word already occupies the complementary index set $\{r + 1, \dots, n\}$, so the same string serves both directions. In the recycling construction X_t is a function of the fiber of U_0 and of $W_{1:t}$, which is (7). The seed stores W^n and a prefix-free encoding of the occupied fiber. Lemma 7 bounds that fiber, and therefore

$$C_n^{\text{causal}}(P, Q) \leq nh + \mathbb{E}[S_m^+] + O(\log n).$$

Lemma 10 (Upper bound).

$$\limsup_{n \rightarrow \infty} \frac{C_n^{\text{causal}}(P, Q) - nh}{\sqrt{n}} \leq \sqrt{\frac{2}{\pi}} |\sigma_P - \sigma_Q|.$$

Proof. Combine the implementation just given with Lemmas 7 and 8. The block length $L = (\log n)^2$ tends to infinity, so the CLT variance identification applies, and $L = o(n^{1/2})$ keeps the edge block inside the $o(\sqrt{n})$ remainder. $\square$

6 The theorem

Theorem 11. *Let P, Q be finite strictly positive laws with $H(P) = H(Q) = h$ and surprisal standard deviations σ_P, σ_Q . Then*

$$C_n^{\text{causal}}(P, Q) = nh + \sqrt{\frac{2}{\pi}} |\sigma_P - \sigma_Q| \sqrt{n} + o(\sqrt{n}). \quad (8)$$

If $\sigma_P = \sigma_Q$, the second term vanishes and the construction of Section 5 gives $C_n^{\text{causal}} = nh + O(\log n)$.

Proof. The case $\Delta = 0$ is the upper bound with a degenerate walk, whose maximum is $O(\log n)$ by the fourth-moment bound, together with the trivial $C_n^{\text{causal}} \geq nh$. For $\Delta > 0$, Lemma 1 and Lemma 6 give the liminf, and Lemma 10 gives the limsup. $\square$

The profile floor of Proposition 2 is the same statement with $\sqrt{2/\pi}$ replaced by $1/\sqrt{2\pi}$. The factor two is the difference between a terminal Gaussian comparison and a running maximum: $\mathbb{E}[(B_1)_+] = \sqrt{1/(2\pi)}$ and $\mathbb{E}[\max_{t \leq 1} B_t] = \sqrt{2/\pi}$.

7 Crash pair

Take $P = (1/2, 1/4, 1/4)$ and $Q = (q, q, 1-2q)$, with $q \in (1/3, 1/2)$ the unique solution of $H(Q) = 3/2$. Then

$$\begin{aligned} q &= 0.4100324281819856\dots, \\ h &= 3/2, \\ \sigma_P^2 &= 1/4, \\ \sigma_Q^2 &= 0.2083473102390734\dots, \\ \sigma_P &= 1/2, \\ \sigma_Q &= 0.4564507752639635\dots \end{aligned}$$

Theorem 11 specializes to

$$C_n^{\text{causal}}(P, Q) = \frac{3}{2}n + 0.0347472540518177\dots \sqrt{n} + o(\sqrt{n}), \quad (9)$$

since

$$\sqrt{\frac{2}{\pi}} |\sigma_P - \sigma_Q| = 0.0347472540518177\dots$$

The two-policy floor is half of this, $0.0173736270259089\dots$. The independent antidiagonal block construction remains a valid, coarser upper bound of order $n^{2/3}$, with leading coefficient $0.145119355153\dots$; Theorem 11 closes that gap.

8 What the argument uses, and what it does not

The atomwise comparison, Lemma 1, is elementary and does not pass through the profile functional $\mathcal{L}$. The square-root barrier for $\mathcal{L}$ proved in the companion note is therefore not a barrier for (2). That barrier says a universal $n^{2/3}$ lower bound cannot be extracted from marginal transcript tails. It does not forbid a sharp constant on the $\sqrt{n}$ scale, which is all that Theorem 11 claims.

The analytic inputs are standard and localized. Well-posedness of (3) is the martingale problem for a uniformly elliptic piecewise constant diffusion coefficient [3]. The passage from marginal convergence to the expectation of the maximum is the envelope inequality $\mathbb{P}(\max_z R_z \leq x) \leq \inf_z \mathbb{P}(R_z \leq x)$, which holds for every coupling, plus Azuma tails for uniform integrability. No identification of the joint law of $(X^{(z)})_z$ is required on the lower side. On the upper side the blocks are independent by construction, so there is no dependence-removal step: $L = (\log n)^2$ is used only to reach the Gaussian variance $(a - b)^2 L$. Prefix-free exactization of U_0 costs $O(\log n)$ bits. The one-way condition (7) is preserved because each output symbol is a function of the seed fiber and the past of W only, and the reversed Q -reads lie off that past.

References

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